

DATA 23700

Project Deliverable 4: Project Repository

See course schedule for the due date.

The project repository is the group's polished, self-contained record of the analysis: a literate-programming analysis notebook, a README documenting the work, and any supporting code. The repository is what a reader would need in order to understand what the group did, why they did it that way, and how to reproduce the results.

Contents

The repository must contain:

- **README** (Markdown, plain text, or PDF): the meta-documentation for the project. See below for required sections.
- **Analysis notebook** (.ipynb or .Rmd): the core analysis, written in a literate-programming style with explicit rationale for every non-trivial analysis and design choice. See below for expectations.
- **Rendered notebook** (.html or .pdf): a version of the analysis notebook with outputs already generated, so graders can read the analysis without having to re-run any code.

Optionally, the repository may also contain:

- **Source code for interactive artifacts or other software:** if the project entails building an interactive display, web app, or other software artifact, the group should include the source code alongside instructions in the README for how to run it locally.
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README

The README should include the following sections:

- **Project summary:** one paragraph stating the motivating problem or question and the group's approach to it. A reader should be able to understand what the project is about from this paragraph alone.
- **Repository structure:** a description of what each file and folder in the repository contains.
- **How to run:** instructions for reproducing the analysis. This should include any environment setup (Python/R version, packages), data acquisition steps if the raw data isn't bundled, and the order in which to run the notebook and any supporting code.

- **Contribution statement:** a description of which group members contributed to which parts of the project. This should be specific enough that a reader can tell that each member did substantive work.

Analysis notebook

The analysis notebook is where the substance of the project lives. It should:

- **Follow a literate-programming style:** prose text and code cells alternate so that the reader can follow the analysis as a narrative. Text should explain what the code does, why the group made the choices they did, and what the outputs reveal.
- **Explain analysis choices:** every nontrivial modeling, wrangling, or statistical decision should be justified in prose. If the group considered alternatives and rejected them, briefly explain why.
- **Explain design choices:** for each key visualization, explain the encoding, color, scale, and other design decisions, and connect these to concepts from the course.
- **Be reproducible:** the code should run start-to-finish in a clean environment, producing the outputs shown in the rendered version. Data should be included in the repository or downloadable via clear instructions in the README.

Groups should aim for a notebook that reads like a technical piece of data journalism or a research paper. The analysis and writing should be woven together, enabling the reader to trace both what the group did and the rationale why.

Submission

Groups submit their project repository as a **zip file to Gradescope** by the deadline. Only one group member needs to submit. That submission is credited to the whole group by entering the names of all group members when submitting to Gradescope.

Fallback for large repositories. If the zip exceeds Gradescope's file size limit, groups may instead submit a link to a Google Drive folder containing the same materials. In that case, upload a placeholder PDF to Gradescope containing the URL, and make sure the folder is shared with view access for the instructor and TAs.

Tips for keeping the zip manageable:

- Clear outputs from the `.ipynb` before zipping, and include a rendered `.html` version separately. Graders read the HTML for the outputs. The notebook re-runs to reproduce them.
- Do not bundle raw datasets larger than a few MB. Instead, include a small data-loading script or a download link in the README.

Grading criteria

The project repository is evaluated on:

- **Quality of analysis:** are the analytical methods appropriate to the data and the question? Are the choices defended in the notebook prose? Is the analysis robust rather than dredging?
- **Quality of visualizations:** do the visualizations apply course principles (encodings, color, scale, etc.) effectively? Are design choices explained and defensible?
- **Quality of documentation:** is the README complete? Is the notebook narrative coherent, and does it read as a piece of technical communication rather than a lab report?
- **Reproducibility:** could a reader with the right environment re-run the analysis from the repository? Are instructions clear?

The project repository may *not* be resubmitted.